ENCN445: Environmental Fluid Mechanics Coastal Engineering Tutorials: Module Four

Problem One: *While stationary, a canoer leaves the cylindrical handle of their paddle hanging in the ocean. They notice that the force on the paddle seems to change over time. After some research they discover that the force acting on the paddle could be modelled with the Morrison equation. The amplitude of the waves when they were paddling was 0.1 m, the period was 6 s, and the phase speed of the wave was 8 m/s. The paddle was 3 cm in diameter and the canoer was in relatively deep water so the depth of the submerged paddle was negligible compared to the fluid depth (i.e. assume that* $\cosh(k(z+h)) = \cosh(kh)$*). The density of the ocean at this location was* $1025\,\mathrm{kg/m^3}$.

A *Use the Morrison equation to separately find the maximum drag force and inertial force acting in the* x *direction on the paddle per unit length. Use a drag coefficient of 1 and an inertial coefficient of 2. The paddle is stationary so you can consider it to be at* $x = 0$.

The Morrison equation is

$$F = F_D + F_I = \frac{1}{2}C_D\rho u|u| + \frac{1}{2}C_I\rho\left(\frac{\pi D^2}{4}\right)\frac{\mathrm{d}u}{\mathrm{d}t}$$

and the velocity of fluid due to a wave is given by

$$u = \frac{gk\eta_0}{\omega}\cos(kx-\omega t)\frac{\cosh(k(z+h))}{\cosh(kh)}.$$

We can ignore the final term involving cosh as we have said that $z+h = h$ and can consider the paddle to be stationary at $x = 0$. This gives the fluid velocity as

$$u = \frac{gk\eta_0}{\omega}\cos(\omega t)$$

The maximum drag force will occur when the velocity is maximum ($\cos(\omega t) = 1$) so

$$u = \frac{gk\eta_0}{\omega} = \frac{g\eta_0}{c} = \frac{9.81\times 0.1}{8} = 0.123\,\mathrm{m/s}.$$

Hence, the maximum drag force per unit length is given by:

$$F_D = \frac{1}{2}C_D\rho D u|u| = \frac{1\times 1025\times 0.03\times 0.123^2}{2} = 0.233\,\mathrm{N/m}$$

To find the maximum inertial force we first need to differentiate the expression for the velocity with respect to time. This gives

$$\frac{\mathrm{d}u}{\mathrm{d}t} = gk\eta_0 \sin(\omega t)$$

The wavenumber can be found from the wave period and phase speed as

$$k = \frac{2\pi}{Tc} = \frac{2\pi}{6 \times 8} = 0.131\,\mathrm{rad/m}.$$

The maximum inertial force will occur when $\sin(\omega t) = 1$ so

$$F_I = \frac{1}{2} C_I \rho \left(\frac{\pi D^2}{4}\right) gk\eta_0 = \frac{1}{2} \times 2 \times 1025 \times \left(\frac{(0.03)^2 \pi}{4}\right) \times 9.81 \times 0.131 \times 0.1 = 0.0931\,\mathrm{N/m}.$$

Problem Two: *Some young New Zealanders have forgone avocado on toast for several decades and are now looking at buying a house. They are looking at a beachfront property but are concerned with the chance of their front lawn flooding during a storm.*

A *They decide to calculate how fast the wind would have to blow to induce a wind setup that would flood their lawn. The lawn is 30 cm above the normal high tide line and the continental shelf out from the house is 40 km wide and 50 m deep. The ocean density is* $1025\,\mathrm{kg/m^3}$*. Use a friction coefficient of* 2×10^{-6}*. Calculate the sustained wind speed needed over the continental shelf to flood the front lawn and advise the potential house buyers how big a risk you think this is.*

The equation for wind setup is

$$\eta_w = h_0 \left(\sqrt{1 + \frac{2\tau_w x}{\rho g h_0^2}} - 1 \right)$$

This can be rearranged to find an expression for the wind stress τ_w needed to produce 30 cm of wind setup at this house:

$$\tau_w = \frac{\rho g h_0^2}{2x} \left[\left(\frac{\eta_w}{h_0} + 1 \right)^2 - 1 \right] = \frac{1025 \times 9.81 \times 50^2}{2 \times 40000} \left[\left(\frac{0.3}{50} + 1 \right)^2 - 1 \right] = 3.78\,\mathrm{N/m^2}.$$

The wind speed required to produce this stress is given by

$$W = \sqrt{\frac{\tau_w}{\rho c_f}} = \sqrt{\frac{3.75}{1025 \times 2 \times 10^{-6}}} = 43.0\,\mathrm{m/s} = 155\,\mathrm{km/h}.$$

A sustained wind of 155 km/h is an extremely high wind speed. It is fair to say that if this wind was blowing the potential owners would have bigger problems than a flooded front lawn!

B *The potential homeowners recognise that wind setup isn't the only component of a storm surge. The baroclinic storm surge is approximately 1 cm for every hPa drop in air*

pressure below 1010 hPa. Assume that a storm has a minimum pressure of 990 hPa and calculate the required windspeed for the total storm surge to overtop the front lawn.
The low pressure is 20 hPa below mean air pressure so we would expect a barcolinic storm surge of approximately 20 cm. Therefore, we only require 10 cm of wind setup for the front lawn to flood. The required wind stress is now given by

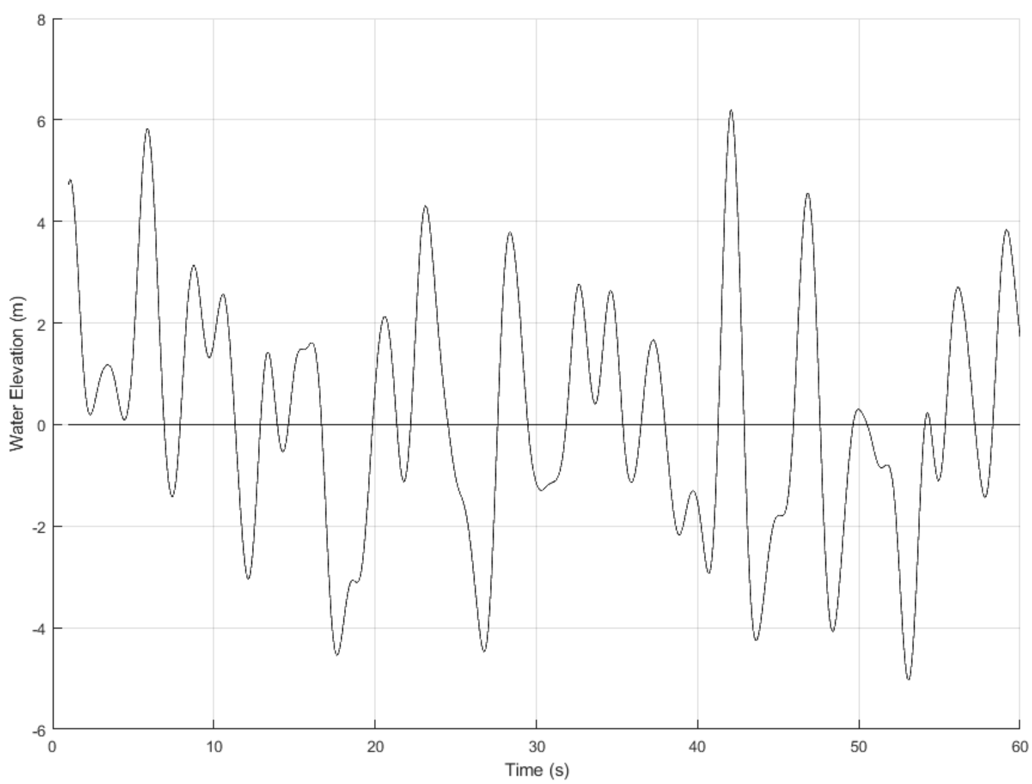
$$\tau_w = \frac{\rho g h_0^2}{2x}\left[\left(\frac{\eta_w}{h_0}+1\right)^2 - 1\right] = \frac{1025 \times 9.81 \times 50^2}{2 \times 40000}\left[\left(\frac{0.10}{50}+1\right)^2 - 1\right] = 1.26\,\mathrm{N/m^2}$$

and the wind velocity is given by

$$W = \sqrt{\frac{\tau_w}{\rho c_f}} = \sqrt{\frac{1.26}{1025 \times 2 \times 10^{-6}}} = 24.8\,\mathrm{m/s} = 89.2\,\mathrm{km/h}$$

In parts of the country sustained wind speeds of 90 km/h are possible and it is likely that the front lawn would flood during a severe storm.

Problem Three:
A client has commisioned a wave survey in preparation for an upcoming project. A representative wave record from typical conditions is shown below.



A *Use an up-crossing analysis to calculate the mean wave height H_z, the mean wave*

period T_z, and the significant wave height H_s
The time of upward zero crossings are

$$t = \{7.9, 12.9, 14.6, 19.9, 22.2, 27.6, 31.8, 36.5, 41.3, 46, 49.7, 54.1, 55.5, 58.3\}\, s.$$

The period, maximum and minimum wave position, and wave height between upward zero crossings is given in the below table.

T (s)	H_{max} (m)	H_{min} (m)	H (m)
5.0	3.1	-3.0	6.1
1.7	1.4	-0.5	1.9
5.3	1.6	-4.5	6.1
2.3	2.1	-1.1	3.2
5.4	4.3	-4.5	8.8
4.2	3.8	-1.3	5.1
4.7	2.7	-1.1	3.8
4.8	1.7	-2.9	4.6
4.7	6.2	-4.2	10.4
3.7	4.6	-4.1	8.7
4.4	0.3	-5.0	5.3
1.4	0.2	-1.1	1.3
2.8	2.7	-1.4	4.1

Where $H = H_{\text{max}} - H_{\text{min}}$. We can now calculate the average of the values in the table to find T_z and H_z:

$$\begin{aligned} T_z &= 3.88\,\text{s} \\ H_z &= 5.34\,\text{m} \end{aligned}$$

There are 13 waves in our wave record so we will use the largest 4 to find the significant wave height H_s. These highest 4 waves have heights of $\{10.4, 8.8, 8.7, 6.1\}$ m. The average of these four heights gives the significant wave height:

$$H_s = 8.5\,\text{m}$$

B *The client is interested in extreme events so wants to know the probability of a wave taller than 15 m in this location. Assuming that waves follow a Rayleigh distribution, calculate the probability of a given wave we be at least 15 m tall.*
Based on the assumption of a Rayleigh distribution we have

$$P(h \geq H) = \exp\left(-2\left(\frac{H}{H_s}\right)^2\right)$$

Using our design height of $H = 15$ m and significant wave height of $H_s = 8.6$ m we find

$$P(h \geq 15) = \exp\left(-2\left(\frac{15}{8.5}\right)^2\right) = 1.97 \times 10^{-3}$$

This is equivalent to one in every 439 waves. Based on our mean wave period of 3.88 s we would thus expect a wave to be taller than 15 m roughly every half hour.

C *Use the spectral width to test how well this wave follows a Rayleigh distribution.*
To calculate the spectral width we also need the mean period between consecutive wave crests. Wave crests occur at:

$$\begin{aligned} t = \{ \; & 3.5, 5.9, 8.7, 10.6, 13.3, 16.1, 18.6, 20.7, 23.1, 28.4, 32.7, 34.6, \\ & 37.3, 39.8, 42.1, 45.1, 46.9, 50.0, 51.7, 54.2, 56.2, 59.2\}\,\mathrm{s}. \end{aligned}$$

The period of each of these waves is

$$\begin{aligned} T = \{ \; & 2.4, 2.8, 1.9, 2.7, 2.8, 2.5, 2.1, 2.4, 5.3, 4.3, 1.9, \\ & 2.7, 2.5, 2.3, 3.0, 1.8, 3.1, 1.7, 2.5, 2.0, 3.0\}\,\mathrm{s}. \end{aligned}$$

giving a mean period

$$T_c = 2.65\,\mathrm{s}$$

From this we can calculate the spectral width as

$$\epsilon = 1 - \left(\frac{T_c}{T_z}\right)^2 = 1 - \left(\frac{2.65}{3.88}\right)^2 = 0.53$$

The closer the spectral width is to 1 the better the approximation that the waves follow a Rayleigh distribution. $\epsilon = 0.53$ is roughly typical for oceanic waves with swell waves estimated to have $\epsilon \approx 0.3$ and storm waves estimates to have $\epsilon \approx \{0.6, 0.8\}$.